

Q.No:8

Question description

You are given an array A of length N which is initialised with 0 . You will be given Q queries of two types:

1 k : set value 1 at index k in array A

2 y : print the smallest index x which is greater than or equal to y and having value 1 . If there is no such index print -1 .

Note: Indexing is 1 based

Constraints

$1 \leq n \leq 10^5$

$1 \leq q \leq 5 * 10^5$

$1 \leq y, k \leq n$

Input Format

First line contains two integers N and Q separated by a space.

The next Q lines contain the type of query (i.e. either a 1 or a 2), then a space, then for type 1 queries integer k and for type 2 queries integer y

Output Format

For each query type 2 , print in new line, the smallest index x which is greater than or equal to y and having value 1 . If there is no such index print -1 .

Explanation for Test case 1

For first query: $2\ 3$, there is no index greater than or equal index 3 , having value -1 , so the answer is -1 .

For second query: $1\ 2$, set value -1 at index 2 .

For third query: $2\ 1$, index 2 is greater than index 1 , having value -1 , so the answer is 2 .

For fourth query: $2\ 3$, there is no index greater than or equal index 3 , having value -1 , so the answer is -1 .

For fifth query: $2\ 2$, index 2 is equal to index 2 , having value -1 , so the answer is 2 .

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 5
2 3
1 2
2 1
2 3
2 2
```

EXPECTED OUTPUT

```
-1
2
-1
2
```

Test Case 2

INPUT (STDIN)

```
5 5
2 1
1 2
2 2
1 2
2 3
```

EXPECTED OUTPUT

```
-1
2
-1
```

Mandatory Test Cases

Test Case 1

KEYWORD

Code:

```
#include <stdio.h>
#include <stdlib.h>

#define MAXQ 500005

long long arr[MAXQ];
int size = 0;
```

```
int compare(const void *a, const void *b)
{
    long long x = *(long long *)a;
    long long y = *(long long *)b;

    if (x < y) return -1;
    if (x > y) return 1;
    return 0;
}

int main()
{
    long long N;
    int Q;

    scanf("%lld %d", &N, &Q);

    int type;
    long long value;
    printf("\nOutput:\n");

    for (int i = 0; i < Q; i++)
    {
        scanf("%d %lld", &type, &value);

        if (type == 1)
        {
            arr[size++] = value;
        }
        else
        {
            long long answer = -1;

            for (int j = 0; j < size; j++)
            {
                if (arr[j] >= value)
                {
                    if (answer == -1 || arr[j] < answer)
                    {
                        answer = arr[j];
                    }
                }
            }
            printf("%lld\n", answer);
        }
    }

    return 0;
}
```

Output:

Test_case: 1

Test Case 1: Graph 1

```
5 5
2 3
1 2
2 1
2 3
2 2
```

Output:

```
-1
2
-1
```

```
2
```

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Test_case: 2

```
5 5
2 1
1 2
2 2
1 2
2 3
```

Output:

```
-1
2
```

```
-1
```

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